

Recurrences, Arithmetic and Geometric Progressions, and the First Examples of Groups

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§1.1 A beginning point: recursive definitions

This note begins with the feeling that recurrence relations look like guessing games. In fact, most elementary recurrences are solved by trying to convert them into one of a few familiar sequences.

The first example is a recurrence of the form

$$a_{n+2} = 5a_{n+1} - 6a_n.$$

If one expects powers, the natural test is to try $a_n = r^n$. Substitution gives

$$r^{n+2} = 5r^{n+1} - 6r^n,$$

so

$$r^2 - 5r + 6 = 0, \quad (r - 2)(r - 3) = 0.$$

Thus the general shape is

$$a_n = A2^n + B3^n.$$

This method is not magic: the recurrence is linear, and exponentials are eigenvectors for the shift operator.

§1.2 Arithmetic progressions

An arithmetic progression is a sequence whose consecutive differences are constant:

$$a_{n+1} - a_n = d.$$

Equivalently,

$$a_n = a_1 + (n - 1)d.$$

The sum of the first n terms is

$$S_n = \frac{n}{2}(a_1 + a_n) = na_1 + \frac{n(n-1)}{2}d.$$

The note emphasizes an important point: if several indices have the same sum, then the corresponding terms of an arithmetic progression also have the same sum. For example, if

$$x_1 + x_2 = y_1 + y_2 = m,$$

then

$$a_{x_1} + a_{x_2} = 2a_1 + (x_1 + x_2 - 2)d = 2a_1 + (m - 2)d = a_{y_1} + a_{y_2}.$$

More generally, if

$$x_1 + \cdots + x_k = y_1 + \cdots + y_k,$$

then

$$a_{x_1} + \cdots + a_{x_k} = a_{y_1} + \cdots + a_{y_k}.$$

This is a small but useful structural observation: arithmetic progressions turn index addition into term addition, up to a constant shift.

§1.3 Determining an arithmetic progression

A typical exercise is: given a_2 and a_5 , find a_1 and d . For instance, if

$$a_2 = 5, \quad a_5 = 11,$$

then

$$a_1 + d = 5, \quad a_1 + 4d = 11.$$

Subtracting gives $3d = 6$, so $d = 2$, and then $a_1 = 3$.

Another exercise gives several sums, for example

$$a_1 + a_2 + a_3 = 24, \quad a_n + a_{n+1} + a_{n+2} = 146,$$

and asks for the number of terms or a particular partial sum. The thought process is always the same: express every term as $a_1 + (k-1)d$, reduce to equations in a_1, d, n , then use the sum formula. The original page highlights the formula

$$S_n = \frac{n}{2}(a_1 + a_n)$$

because it often avoids writing every term.

§1.4 Geometric progressions

A geometric progression is a sequence whose ratios are constant:

$$\frac{a_{n+1}}{a_n} = q.$$

Hence

$$a_n = a_1 q^{n-1}.$$

If $q \neq 1$, then

$$S_n = a_1 \frac{q^n - 1}{q - 1}.$$

If $q = 1$, then

$$S_n = na_1.$$

The note gives the derivation by subtracting:

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1},$$

$$qS_n = a_1 q + a_1 q^2 + \cdots + a_1 q^n,$$

so

$$(q-1)S_n = a_1(q^n - 1).$$

This derivation is better than memorization because it also explains why the case $q = 1$ must be separated.

§1.5 Converting recurrences

The page contains the example

$$a_{n+1} = 2a_n + 1.$$

There are two standard ways to solve it.

First, shift the sequence by a constant. Try $b_n = a_n + c$, and choose c so that the recurrence becomes homogeneous. Since

$$a_{n+1} + 1 = 2(a_n + 1),$$

we let $b_n = a_n + 1$. Then

$$b_{n+1} = 2b_n,$$

so $b_n = 2^{n-1}b_1$. If $a_1 = 1$, then $b_1 = 2$, and

$$a_n = 2^n - 1.$$

Second, one may divide by a power of 2. This often works for recurrences of the form

$$a_{n+1} = qa_n + r_n.$$

The goal is to make the left side telescope.

§1.6 A factorial recurrence

Another example is

$$a_n = na_{n-1} + 1, \quad a_1 = 1.$$

Dividing by $n!$ gives

$$\frac{a_n}{n!} = \frac{a_{n-1}}{(n-1)!} + \frac{1}{n!}.$$

Thus if

$$b_n = \frac{a_n}{n!},$$

then

$$b_n = b_{n-1} + \frac{1}{n!}.$$

Therefore

$$a_n = n! \left(1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} \right).$$

The point is not the final formula alone. The point is that multiplication by n suggests normalizing by $n!$.

§1.7 Groups

The second half of the note switches to group theory. A group is a set G with a binary operation $*$ satisfying:

- (i) associativity: $(a * b) * c = a * (b * c)$;
- (ii) identity: there is 1_G such that $1_G * g = g * 1_G = g$;
- (iii) inverses: for each $g \in G$, there is g^{-1} with $gg^{-1} = g^{-1}g = 1_G$.

Commutativity is not required. If it holds, the group is abelian.

Standard examples include:

$$(\mathbb{Z}, +), \quad \mathbb{Z}/n\mathbb{Z}, \quad S^1 = \{z \in \mathbb{C} : |z| = 1\}.$$

The note explicitly warns that closure is already included in the phrase “binary operation on G ”: if $*$: $G \times G \rightarrow G$, then $g * h \in G$ automatically.

§1.8 Symmetric and dihedral groups

The symmetric group S_n consists of all permutations of n objects. Composition is the group operation. The note writes permutations in cycle notation and emphasizes that the order of composition matters.

The dihedral group D_{2n} is the symmetry group of a regular n -gon. It contains rotations and reflections. It has presentation

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1, \, srs = r^{-1} \rangle.$$

For $n = 5$, for example, r is a rotation by 72° , and s is a reflection. The relation $srs = r^{-1}$ says that reflecting reverses the direction of rotation.

§1.9 Products and isomorphisms

If G, H are groups, their product is

$$G \times H = \{(g, h) : g \in G, \, h \in H\},$$

with operation

$$(g_1, h_1)(g_2, h_2) = (g_1g_2, h_1h_2).$$

The inverse is

$$(g, h)^{-1} = (g^{-1}, h^{-1}).$$

Two groups are isomorphic if there is a bijection $\varphi : G \rightarrow H$ such that

$$\varphi(g_1g_2) = \varphi(g_1)\varphi(g_2).$$

An isomorphism means that the two groups have the same structure with different labels.

§1.10 Linear recurrences

A first-order linear recurrence has the form

$$a_{n+1} = ra_n + b.$$

If $re1$, look for a fixed point L satisfying $L = rL + b$. Then subtract it:

$$a_{n+1} - L = r(a_n - L).$$

So

$$a_n - L = r^n(a_0 - L).$$

This method explains why geometric progressions appear inside many non-homogeneous recurrences.

§1.11 Groups from repeated operations

The note's first group examples should be read as abstraction from repeated operations. Integers under addition, nonzero rationals under multiplication, rotations of a polygon, and permutations are all systems where operations can be composed and undone. The group axioms isolate exactly that pattern: associativity, identity, and inverses.

§1.12 Why the pattern matters

This note is valuable because it places recurrences and groups next to each other. A recurrence is about repeatedly applying an operation; a group is about operations that can be composed and undone. Both topics are early appearances of the same general idea: mathematics often studies not isolated objects, but transformations and their iterates.

§1.13 Recurrences as iteration of maps

A recurrence $a_{n+1} = F(a_n)$ is iteration of a map. Arithmetic progressions iterate a translation, and geometric progressions iterate a scaling. Their closed forms are orbit formulas:

$$a_n = a_1 + (n-1)d, \quad a_n = a_1 q^{n-1}.$$

§1.14 First-order linear recurrences

A recurrence

$$a_{n+1} = ra_n + b$$

has a fixed point $a_* = b/(1-r)$ when $r \neq 1$. Subtracting the fixed point gives

$$a_{n+1} - a_* = r(a_n - a_*).$$

This turns an affine recurrence into a geometric progression.

§1.15 Groups behind repeated operations

Groups formalize reversible repeated operations. Integers under addition record repeated translation; nonzero real numbers under multiplication record repeated scaling. The recurrence examples are therefore early examples of group actions on a set.

§1.16 Sequences as orbits of operations

Sequences become clearer as dynamics. Arithmetic and geometric progressions are translation and scaling orbits, affine recurrences become geometric after shifting to a fixed point, and groups record the reversible operations behind iteration.